MIL-W-36451
27 October 1965

MILITARY SPECIFICATION

WASHER-STERILIZER, SURGICAL INSTRUMENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers one type of pressure washer-sterilizer, suitable for washing and sterilizing surgical instruments.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

CC-M-636c	Motors, Fractional Horsepower, Alternating Current.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated Triplewall.
PPP-C-843	Cushioning, Material, Cellulosic.
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging.

FSC 6530

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Military

MIL-C-7769	Cushioning, Material, Uncompressed Bound Fiberboard - Vaporproof for Packaging.
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-P-116	Preservation, Methods of

STANDARDS

Federal

FED-STD-151	Metals, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

2.2 Other publications.

(American Society for Testing Materials) ASTM

A 265	Spec. for Nickel and Nickel-Base Alloy Clad Steel Plate.
A 276	Spec. for Hot-Rolled and Cold-Finished Stainless and Heat Resisting Steel Bars.
A 340	Terms, Symbols and Conversion Factors Relating to Magnetic Testing.
B 143	Spec. for Tin Bronze and Leaded Tin Bronze Castings.
B 162	Spec. for Nickel Plate, Sheet and Strip.
D 735	Elastomer Compounds for Automotive Applications.

Application for copies should be addressed to:
American Society for Testing Materials
1916 Race Street
Philadelphia, Pennsylvania 19103

(American Society of Mechanical Engineers) ASME

SA 255	Method of End-Quench Test for Hardenability of Steel.
SA 285	Spec. for Low and Intermediate Tensile Strength Carbon-Steel Plates of Flange and Firebox Qualities (Plates 2 in. and under in thickness)
SB 62	Spec. for Composition Bronze or Ounce Metal Castings.
SB 164	Spec. for Nickel-Copper Alloy Rod and Bar.
Section B	Boiler and Pressure Vessels Code (Unfired Pressure Vessels).

Application for copies should be addressed to:
American Society of Mechanical Engineers
345 East 47 Street
New York 17, New York

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(National Fire Protection Association) NFPA
No. 56 Code for the Use of Flammable Anesthetics.

Application for copies should be addressed to:
National Fire Protection Association
60 Batterymarch Street
Boston, Mass. 62110

(Underwriters' Laboratories, Inc.) UL
499 Standards for Safety, Electric Heating Appliances.

Application for copies should be addressed to:
Underwriters' Laboratories, Inc.
207 East Ohio Street
Chicago, Illinois 60611

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3. Preproduction sample. Unless otherwise specified (see 6.), prior to the start of production of the contract quantity, one preproduction sample of the washer-sterilizer shall be assembled, completely finished, and tested by the contractor. The sample shall be submitted, together with the contractor's inspection and test reports, to the contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Further production of the item by the contractor, prior to this approval, shall be at the contractor's risk. Accepted preproduction sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the preproduction sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.1.1 Fire and casualty hazards.

3.1.2 The contractor shall submit to the contracting agency proof, prior to delivery, that the equipment he will supply under this specification conforms to the applicable requirements of Underwriters' Laboratories, Inc. Standard UL 499 and the National Fire Protection Association No. 56. Such proof shall consist of a laboratory report or listing by the Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories shall be acceptable as evidence that the equipment conforms to the above. A copy of each laboratory report shall be furnished to the contracting officer.

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3.1.3 In lieu of the label or listing by the Underwriters' Laboratories, Inc., the contractor may submit a laboratory report by a nationally recognized testing agency adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp or label to the article, or the issuance of a certificate. The contractor shall submit the laboratory report which shall include test results and methods used to determine compliance, to the contracting officer for approval. The Government reserves the right to require a facilities report showing the qualifications of the testing laboratory and to conduct a facilities survey to determine its acceptability.

3.1.4 Compliance with the above requirements regarding fire and casualty hazards, does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Material.

3.2.1 Nickel. Nickel shall have a thickness of 0.0250 inch, a polished satin finish and shall conform to ASTM B 162.

3.2.2 Nickel-copper alloy. The alloy shall have a thickness of 0.250 inch, a polished satin finish and shall conform to ASME SB 164.

3.2.3 Flange quality steel. The steel shall have a thickness of 0.250 inch, a mill finish and shall conform to ASME SA 285, Grade C - Hot Rolled.

3.2.4 Nickel clad steel. Nickel clad steel shall conform to ASTM A 265.

3.2.5 Corrosion-resistant steel. The CRS shall be of 0.625 inch x 0.750 inch stock, have a polished satin finish and conform to ASTM A 276 (Type 416).

3.2.6 Cast brass. The brass shall have a thickness of 0.375 inch, a chromium whitened finish and shall conform to ASME SB 62.

3.2.7 Cast bronze. The bronze shall have a thickness to meet ASME requirements, a satin chromium finish and shall conform to ASTM B 143.

3.2.8 Insulation. The insulation shall be glass fiber with a density of 0.750 pound per cubic foot, a thickness of one inch, and a mill finish.

3.2.9 Elastomer Compound. The gasket shall be elastomer compound, grade SG 615 A1 Z1 Z2, have a thickness of 0.360 inch, a raw finish and shall conform to ASTM D 735.

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3.2.10 Fasteners. Rivets, bolts, nuts, studs, spacers, and metals used for welding shall be corrosion-resistant steel or nickel-copper alloy where these metals are joined to each other or to other metals. Materials used to join metals other than corrosion-resistant steel or nickel-copper alloy shall be such as to prevent galvanic corrosion.

3.3 Style and design. The washer-sterilizer shall be a single door, cabinet mounted unit. The sterilizing chamber shall be automatically controlled and direct steam heated.

3.4 Construction. The washer-sterilizer shall consist of a sterilizing chamber complete with door and tray arrangement. It shall also include a heating system, an exhausting system, piping, valves, fittings, and a cabinet. Details of construction, materials and testing of the washer-sterilizer shall comply with the requirements in Section B of the ASME Boiler and Pressure Vessels Code (Unfired Pressure Vessels). Each washer-sterilizer shall be designed for the pressure specified hereinafter and shall bear the stamp of the ASME and the National Board of Boiler and Pressure Inspectors.

3.4.1 Sterilizing chamber. The chamber shall consist of a single shell, a backhead and end frame assembly. It shall be square, horizontal and 16 inches by 16 inches by 24 inches in size.

3.4.1.1 Shell. The shell shall be of all welded construction and consist of flange quality steel having a tensile strength of 55,000 to 65,000 pounds per square inch. The steel shall conform to 3.2.3 of this specification. The steel forming the interior surface of the shell shall be clad with pure nickel or nickel-copper alloy bonded to the steel. The nickel-copper alloy and the nickel shall conform to 3.2.2 and 3.2.1 respectively.

3.4.1.2 Backhead. The backhead shall be of the same material as the shell, and shall be fusion welded in place.

3.4.1.3 End frame. The end frame assembly shall consist of a head-ring and door frame made of nickel-copper alloy conforming to 3.2.2. The door frame shall receive, support and conceal the ends of the door locking arms when the door is in the tightened position. The head-ring or door frame surface which makes contact with the door gasket shall be smooth.

3.4.2 Door. The door shall be of the radial locking arm type and made of cast bronze as specified in 3.2.6. It shall be manually operated, and designed and constructed to fit over the opening of the sterilizing chamber. A groove shall be provided in the door for the sealing gasket.

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3.4.2.1 Hinge. The hinge shall be made of cast bronze as specified in 3.2.6. It shall be located on the right, and be reversible in the field without the need for welding, additional parts or machine work. It shall be fitted with anti-friction bearings and be provided with a vertical and horizontal adjustment to insure central closure of the door.

3.4.2.2 Lock. The door shall have at least a dual safety-lock consisting of:

a. An inner lock activated by internal pressure and unlocked only by the exhaust of chamber pressure.

b. An external mechanical lock which shall prevent disengagement of the locking arms until the chamber pressure has been exhausted.

The door lock shall be designed and constructed so that the door may be tightened but not loosened with pressure in the chamber. The door lock shall be controlled by a centrally located metal handwheel covered with low-heat-conducting, baked-on vitreous enamel. As an alternate the centrally located door lock control shall be provided with not less than three non-heat conducting, phenolic, thermosetting plastic radial handles with reinforcing metal inserts.

3.4.2.3 Gasket. The sealing gasket used for the door shall be a continuous piece of material conforming to 3.2.8.

3.4.2.4 Insulation. The exterior of the sterilizing chamber, except for the door, shall be covered with fiberglass insulation as specified in 3.2.7. The insulation shall be backed with aluminum foil and held in place with thermosetting tape.

3.4.2.5 Trays. Each unit shall have three trays and a file drawer suspension type rack. Each tray shall have a perforated bottom with not more than 12 openings per square inch and nominal dimensions of 21 inches long by 10 inches wide by 3 inches deep. As an alternate, each tray shall have a nominal width of 13-5/8 inches. Each tray shall have two suitable carrying handles securely attached. The trays and rack shall be made of nickel-copper alloy as specified in 3.2.2. As an alternate, the trays and rack shall be made of type 304 corrosion-resistant steel with a thickness of 0.025 inch.

3.4.3 Cabinet. The washer-sterilizer shall be supported by a rust-proofed angle steel framework having four cadmium plated adjustable floor level supports. To form a cabinet, the framework shall be enclosed on top, front and sides with panels made of corrosion-resistant steel conforming to ASTM A 177 (Type 302) and having a thickness of not less than 0.037 inch. The panels shall have a satin finish and the front panel shall have openings through which the door frame of the sterilizer, the door, and all necessary valve handwheels and gauges shall extend. The top of the cabinet shall have a sliding access panel easily removable for servicing.

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3.4.4 Operational components. The washer-sterilizer shall be complete with all of the necessary piping, valves and fittings required for the control of the water and steam supply and for draining the unit. All pipes, valves, and fittings shall be adequately secured in position to prevent them from vibrating, and also to reduce strain where connected to sheet metal.

3.4.4.1 Piping. The piping shall be rigid, seamless brass where it serves as support for components, and annealed copper in other locations. The piping shall be for direct connection to the building steam supply, water supply and return lines, and shall operate satisfactorily on steam pressure of 40 to 80 pounds per square inch at the fixture. The piping shall be designed and constructed to prevent the back flow of polluted water or waste into the water supply system or the sterilizer.

3.4.4.2 Valves. The valves furnished with each sterilizer shall be made up of at least the following types:

(a) Cam operated valves.

(b) Safety valve.

(c) Pressure regulator valve.

3.4.4.3 Cam operated valves. Cam operated valves shall be of bronze with renewable nickel-copper alloy seats as specified in 3.2.2 and polytetrafluoroethylene discs. Valves shall be leakproof when tested at 100 pounds air pressure with valve body submerged in water. Manually operated valves shall be of the globe or angle body type and have low-heat conducting non-loosening handles, easily removable and replaceable but rigidly secured to valve stem. Durable letters with indicative coloring on face of handle shall show the purpose of the valve, e.g., "water", "steam", etc. Valves shall be mounted in an accessible place and where necessary, be provided with extension stems to permit operation from the front of the sterilizer. Wall sleeves and collars shall be furnished as necessary. Valves shall open when handle is turned counter-clockwise.

3.4.4.4 Safety valve. Each sterilizer shall have a safety valve which shall be approved by and bear the stamp of the ASME. The safety valve shall be locked so that it may be unlocked only with tools. The valve shall be set to operate at 33 pounds per square inch and shall not allow the pressure in the chamber to rise above 33 pounds per square inch. The valve shall blow down at least 2 pounds per square inch, but not below 28 pounds per square inch, before shutting off.

3.4.4.5 Pressure regulator valve. Each washer-sterilizer shall have an automatic steam pressure regulator valve. This valve shall maintain the pressure within the chamber at a level sufficient to achieve a sterilizing temperature of 270° F to 272° F as required during each operating cycle.

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3.4.4.6 Traps. Each thermostatic steam trap shall be pressure-balanced and able to withstand a maximum pressure of 60 pounds per square inch. It shall discharge condensate and air during the heating and sterilizing periods. The trap shall close against steam, be compensated for pressure and temperature and have renewable element, seat and gasket.

3.4.4.7 Water supply. The washer-sterilizer shall be provided with the following means for prevention of backflow and contamination of either the source of water supply or contents of the sterilizer:

(1) An air-brake or air-intake which shall admit water into the chamber but shall provide against syphoning of the contents of the sterilizer back into the water supply line.

(2) A device that shall prevent the leakage of water into the sterilizer through a leaking valve after the sterilizer has been filled and the valve shut.

3.5 Performance. The control and operation of the washer-sterilizer shall be completely automatic for both the wash-sterilize and the sterilize only cycles. Provision shall be made so that the unit may be manually operated, by means of a single control handle, in the event of an electrical power failure. The automatic operation shall be started by activating a pushbotton or selector switch, which shall be located in the front panel above the door frame. When the door is closed and the control is activated, the automatic operation of the washer-sterilizer shall consist of either of the following cycles:

(a) Wash and sterilize.

1. The chamber shall be filled with cold water and steam injected to loosen gross soil on the load.

2. The load shall be soaked by means of a variable soak period ranging from 0 to 15 minutes.

3. Steam shall be injected into the chamber until the wash water is heated to a nominal temperature of 150°F. At the same time, the articles comprising the load shall be washed by forceful steam agitation of the water.

4. The water shall be drained from the chamber.

5. Steam shall be admitted to sterilize the load at 270° F. At this time, the three minute timing period shall begin.

6. At the end of the three minute period, the chamber steam pressure shall be exhausted.

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7. An audible signal shall be emitted when the chamber pressure reaches approximately atmospheric, to indicate completion of the cycle.

(b) Sterilizing only.

1. The chamber shall be charged with steam at not less than 270° F for the selected period of time (3 minutes or 7-10 minutes).

2. The steam in the chamber shall be exhausted to the condenser and then to the waste line.

3. When the chamber pressure reaches approximately atmospheric, an audible signal shall be emitted to indicate completion of the cycle.

The unit shall be safe for use in areas where explosive medical gases are administered, such as surgery.

3.5.1 Heating system. The washer-sterilizer shall be heated directly from the steam line. It shall operate satisfactorily on steam supply pressures from 40 to 80 pounds per square inch at the fixture. A steam strainer shall be provided in the steam supply line, which shall effectively clean the steam of solid matter and be readily removed for cleaning. The steam strainer wire shall be nickel-copper alloy as specified in 3.2.2. Heating shall be accomplished by means of a muffled steam jet heater. The heater shall be designed and constructed so that when heating it will induce circulation of the water upward and toward an edge screened overflow to direct gross soil into the waste line. No heating coils shall be used.

3.5.2 Exhaust system. The exhaust system shall contain a condenser to condense steam exhaust and discharge chamber condensate for disposal to the waste system.

3.5.3 Automatic control. The automatic control unit shall be motor driven and cam operated. After activation of the unit by means of a switch or push-button, no further attention shall be required by the operator. The timing shall be temperature activated by means of a sensing bulb located in the chamber drain line. The unit shall be manually operable in the event of electric power failure and shall comply with the requirements of the NFPA Code No. 56. The unit shall be designed for operation on 120 volt, 60 cycle, AC.

3.5.3.1 Timer. The automatic control unit shall contain a synchronous motor-driven timer with the sterilization period adjustable by means of a single knob for any desired exposure time within the range of 0 to at least 30 minutes. The timer shall cycle only if operating temperature is present in the chamber. The timer shall automatically reset to its starting point, if during the sterilizing period the chamber temperature should fall below the established minimum, or the electric power should fail.

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3.5.4 Motor. The motor shall conform to Federal Specification CC-M-636c, except that radio interference suppression shall not be required. The motor shall be suitable for operation on 110 volt, single phase, 60 cycle, AC. The motor shall have an open frame, class A insulation, and a continuous duty classification based on 50° C ambient temperature of reference. The motor shall be capable of performing the required functions and shall not exceed its nameplate rating when related equipment is operating at specified capacity.

3.5.5 Thermometer. The sterilizer shall be provided with a dial type temperature indicating instrument. The calibrated graduations shall be from not more than 230° F to at least 275° F. The thermometer shall be easily read and be accurate to ± 1° F within the calibrated range. One sixteenth (1/16) inch linear scale distance shall correspond to not more than 4° F. The temperature sensing bulb shall be located in the coolest part of the chamber. The thermometer shall be mounted in the front panel or in the sterilizer door.

3.5.6 Steam pressure gauge. A steam pressure gauge shall be supplied to indicate the pressure in the chamber. The gauge shall be calibrated from 30 pounds per square inch to 60 pounds per square inch and shall be accurate to within 2 pounds of gauge. The gauge shall be mounted in the front panel above the door.

3.5.7 Fuse. The washer-sterilizer shall be adequately fused to prevent damage to the electrical components.

3.6 Finish. All exterior corrosion-resistant steel surfaces shall be finished as specified. All plated surfaces shall be given a mirror finish. Welds in the nickel-copper alloy shell shall shall be blended with the adjacent mill finish on the sheet.

3.7 Identification marking. Each sterilizer shall be fitted, in a conspicuous location, with a metal nameplate. The nameplate shall contain the following data:

Name or registered trade-mark of the manufacturer.
Manufacturer's type designation.
Serial number.
Voltage and kilowatt rating.
Steam supply line pressure range.
Water supply line pressure range.
Stock number.
Contract or purchase order number.
U.S.

Marking secured by adhesives shall not be used.

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3.8 Repair parts. Each sterilizer shall be provided with three spare pilot lamps.

3.9 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. This information shall specifically include instructions and drawings for plumbing, power wiring, structural modifications, and special support devices necessary for roughing-in preparations prior to installation of the unit. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. One set of the service data, as indicated above, shall be placed in a sealed, waterproof envelope and affixed securely to the outside of the exterior container. The envelope shall be secured to the exterior container by use of metal ties, nails, or similar means, and shall be located at the end of the exterior container, preferably, when space permits, at the end bearing the markings required by MIL-STD-129. In cases where complete service data is too bulky to be conveniently attached to the outside of the exterior container, the contents of the exterior envelope may be limited to information necessary for roughing-in preparations, and a note shall be made on the exterior envelope which shall indicate where the remainder of the first set of service data will be located.

When two or more containers are required for packing a disassembled item (components of a single stock number), set markings shall be applied in accordance with MIL-STD-129. The envelope shall be affixed to the container designated as No. 1 of the set. The second set and, if necessary, the remainder of the first set of service data, shall be attached to the unit inside the exterior container. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

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3.10 Workmanship. The workmanship, finish, fit, safety, and details of apparatus shall be first class in every respect. Sterilizers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that the supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.1 Examination. Sampling for examination shall be conducted in accordance with MIL-STD-105

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total (major and minor combined) defects	II	4.0

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4.2.2 For testing.

	Characteristic	Reqt	Test Procedure	Inspection level
Sterilizer	Safety locking device	3.4.1.5.2	4.4.2	100%
	Door hinge adjustment	3.4.1.5.1	4.4.2	100%
	Steam tightness	3.4.1.2	4.4.3	100%
	Thermostatic trap functions	3.4.3.3	4.4.2	100%
	Pressure and temperature maintenance	3.4.3.2.3	4.4.2	100%
	Airbreak functions	3.4.3.4	4.4.2	100%
	Automatic Wash-sterilize cycle	3.4.4	4.4.2 4.4.6.1	100%
	Automatic 3-minute Sterilize cycle	3.4.4	4.4.2 4.4.6.1	100%
	Automatic 10-minute Sterilize cycle	3.4.4	4.4.2 4.4.6.1	100%
	Manual operation	3.4.4	4.4.2	100%
	Dielectric strength	3.4.4	4.4.5	100%
Safety valve	Excess pressure	3.4.3.2.2	4.4.3	100%
Valves	Hydrostatic pressure	3.4.3.2.1	4.4.4	S-3
Thermometer	Accuracy	3.4.4.5	4.4.6.2	S-3
Steam pressure gauge	Accuracy	3.4.4.6	4.4.6.3	S-3
Timing device	Accuracy	3.4.4.3.1	4.4.6.1	S-3

4.2.3 Raw materials. Four ounces of each lot of metal used in the construction of the washer-sterilizer shall be selected for chemical analysis.

4.3 Examination. The washer-sterilizer shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects *

Major

101 Any component part missing.
102 Exposed edges not free of burs and sharpness.
103 Any component not free of cracks, ruptures, and tears.
104 Finish not free of poor adhesion i.e., blistering, checking, or peeling.
105 Any weld omitted, not fused, not free of burns, cracks, and fractures.
106 Any threaded part stripped, or does not function smoothly.
107 Any unit does not bear stamp of ASME and the National Board of Boiler and Pressure Inspectors.
108 Access panel not easily removable, does not allow access for convenient servicing of any enclosed component.
109 Any part on control panel not properly identified.
110 Identification marking missing, incomplete, illegible or not properly secured.
111 Gasket contacting surface of end frame not smooth.
112 Leaks in water, steam or drain lines.
113 Insulation not held in place as specified.
114 Door hinge does not provide vertical and lateral adjustment as specified.
115 Gasket not properly seated in door.

Minor

201 Service data incomplete or missing.
202 Any pipe, valve, or fittings not adequately secured in position.
203 Threads at joints not as specified.
204 Exterior cabinet does not have uniform finish free of scratches or other marks which detract from its appearance.
205 Welds in shell not polished to blend with adjacent surfaces.
206 Gages, dials or controls not neatly flush mounted on front surface of cabinet.
207 Any dimension not as specified.
208 Exterior surfaces of shell not uniformly insulated.
209 Drawer suspension supports do not operate smoothly.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The trays shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with method 111.1 or 112.1 of Fed. Test Method Std. No. 151.

4.4.2 Operation test. The washer-sterilizer shall be connected to a steam line at 50 psi, a water line at 25 psi, and a power source, and operated in accordance with manufacturer's instructions. The washer-sterilizer shall first be operated on the automatic wash-sterilize cycle. The report shall include the following observations:

Fill time: Nominal 4 minutes
Wash time: Nominal 10 minutes
Wash temperature at end of cycle where wash water is heated: 140° F.
Time to reach 270° F. and 27 psi after unit has finished draining: 6 minutes
Time for sterilization: 3 minutes -0 +1 minute
Exhaust time: 1 minute
Total time for complete cycle: Nominal 25 minutes

The unit shall then be operated as a sterilizer by turning the selector switch or setting the timer to a 3 minute sterilize cycle and then to a 10 minute sterilize cycle.

4.4.3 Excess pressure. Pressures in excess of 30 psi shall be introduced into the washer-sterilizer. The report shall include the following observations:

Pressure at which the safety valve releases:
Pressure at which the safety valve blows down:
The safety valve continues/does not continue to reseat itself after ten repeated ventings of steam.

With the safety valve removed, the washer-sterilizer shall be subjected to a test steam pressure of 42 psi.

The door is/is not steamtight.

4.4.4 Hydrostatic pressure-valves. Valves shall be subjected to an air pressure of 100 psi. The valves shall be submerged in water, and observed for evidence of leakage.

4.4.5 Dielectric strength. All current-carrying parts shall withstand, without failure, a dielectric test of 900 volts, 60 cycles, single phase, AC, supplied for 1 minute between the exposed noncurrent-carrying parts and the current-carrying parts, with the unit at maximum operating temperature attained in normal use.

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4.4.6 Accuracy.

4.4.6.1 Timing device. Timing device shall be measured for accuracy using equipment accurate to the nearest 0.1 minute.

4.4.6.2 Thermometer. Thermometer shall be measured for accuracy using equipment accurate to the nearest 1/2° F.

4.4.6.3 Steam pressure gauge. Steam pressure gauge shall be measured for accuracy using equipment accurate to the nearest 1/4 psi.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) sterilizer, as specified, constitutes one unit of issue.

5.1.2.1 Unit preparation. The sterilizer shall be adequately protected by cushioning with corrugated pads, cellulosic cushioning material conforming to Specification PPP-C-843 and/or type II bound fiber conforming to Specification MIL-C-7769 to prevent marring, scratching, or other damage. All switch boxes, outlets, connections and drain line openings shall be sealed with PPP-T-60, type II, class 1 tape. Indicators, gauges and dials shall be protected with one of the above cushioning media and secured with PPP-T-60 tape.

5.1.2.2 Trays and racks. Secure the trays and racks in position by tying, blocking, or bracing to prevent movement during transit.

5.1.2.3 Instruction books and parts list. Instruction books and parts list shall be packaged in accordance with MIL-P-116, method 1C1 and secured to the sterilizer in a protected location.

5.1.3 Level C. Sterilizers shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit prepared as specified in 5.1.2 shall be overpacked in an exterior container designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-640, class 2, grade A.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, the requirements of paragraph 5.5.16.3 shall apply.

6. NOTES

6.1 Intended use. Pressure washer-sterilizer, for washing and sterilizing surgical instruments, permanently installed.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5. and 5.).

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

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